

```
import java.util.Scanner;

public class BinarySearch {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input array
        System.out.println("Enter the number of elements in the array: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the elements of the array (sorted in ascending order): ");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        // Input element to search
        System.out.println("Enter the element to search: ");
        int key = sc.nextInt();

        // Search for the element using binary search
        int index = binarySearch(arr, key);

        // Display result
        if (index == -1) {
            System.out.println("Element not found!");
        } else {
            System.out.println("Element found at index: " + index);
        }

        sc.close();
    }

    // Binary search function
    public static int binarySearch(int[] arr, int key) {
```

```
int low = 0;
int high = arr.length - 1;
while (low <= high) {
    int mid = (low + high) / 2;
    if (arr[mid] == key) {
        return mid;
    } else if (arr[mid] < key) {
        low = mid + 1;
    } else {
        high = mid - 1;
    }
}
return -1;
}
```